

Ethan Chan

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EDUCATION

University of California, Riverside

B.S. in Statistics (Statistical Computing Concentration)

Riverside, CA

Expected June 2026

PROFESSIONAL EXPERIENCE

Research Assistant - Graph Unlearning with LLM Agents

June 2025 - August

University of California, Riverside

- Independently adapted existing graph unlearning codebase, integrating Llama Maverick 4 to select which graph nodes/edges to remove while resolving API compatibility and integration challenges.
- Scaled system from small graphs with 1K nodes to large datasets with 169K nodes and 1.2M edges, building chunking and checkpointing systems to enable reliable experiments on massive datasets.
- Extended pipeline to handle 15+ different graph types, implemented 6 ranking methods for edge selection, and added comprehensive error handling to process large edge datasets efficiently.

Research Assistant - Multi-Omics on Rice

January 2026 - Present

University of California, Riverside

- Currently conducting research on benchmarking multi-omics phenotype prediction.

GrandMarc at University Village

April 2024 - September 2025

Leasing Agent

Riverside, CA

- Assisted with resident relations, leasing documentation, and daily package distribution for an 800+ tenant community, improving satisfaction and operational accuracy.
- Utilized Excel and mail merge to send personalized move-in information to incoming residents and conducted outreach to prospective tenants, supporting successful lease signings.

PROJECTS

Flight Delay Analysis

April - June 2025

- Conducted exploratory data analysis (EDA) on 330,000+ NYC flights (nycflights13, dplyr, ggplot2), engineered seasonal features, and developed linear regression models ($\text{dep_delay} \sim \text{season} * \text{carrier}$) validated with 10-fold cross-validation (RMSE $\sim$ 40 min).
- Identified that pre-dawn departures and spring travel incur the highest delays; found ExpressJet to be worst in winter and Delta to be the most punctual.

Sleep Cycle Productivity Analysis

March 2025

- Analyzed a synthetic sleep dataset (5,000 observations) to explore how sleep duration and quality relate to productivity across age groups, producing comparative summaries and visualizations (bar charts, radar plots, scatterplots) in Excel.
- Applied ANOVA and t-tests to reveal significant productivity differences between sleep groups ($p = 0.033$) and found a moderate correlation between sleep duration and productivity ($r = 0.66$), highlighting the impact of healthy sleep patterns.

Business Analysis

September - December 2024

- Conducted EDA on $\sim$ 8,500 Sephora products and over 1M reviews using R (tidyverse, tidytext, ggplot2) to examine relationships between product sentiment, quality ratings, and engagement metrics.
- Performed sentiment analysis on review text and integrated results into a regression model ($R^2 \approx 0.89$), revealing products with high visibility but low satisfaction and highlighting gaps between consumer attention and perceived value.

Technical Skills

Programming Languages: Python, C++, R

AI/ML Tools: NumPy, pandas, PyTorch, HuggingFace, Excel